Failure-Aware Episodic Memory for Autonomous Penetration-Testing Agents

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Project Description

Recent advances in large language models (LLMs) have enabled autonomous AI agents for cybersecurity tasks such as vulnerability analysis, penetration testing, and attack simulation [1, 2]. However, current agents struggle with long-horizon tasks and often fail to learn effectively from unsuccessful actions [5]. Prior work has therefore proposed episodic memory and reflection mechanisms to help agents retain past observations, failures, and outcomes for use in future decision-making [3, 4].

In penetration-testing environments, failures such as unsuccessful exploit attempts, revisiting dead-end attack paths, or getting trapped in ineffective exploration loops significantly reduce agent efficiency [2, 5]. In realistic cybersecurity settings, defenders may intentionally deploy deceptive mechanisms such as decoy vulnerabilities or honeypots to waste attacker resources and trap automated agents into repeated failures. These challenges suggest that remembering and avoiding prior failures may be as important as remembering successful attack trajectories.

This project investigates whether failure-aware episodic memory can improve the efficiency and robustness of autonomous penetration-testing agents. Episodic memory refers to the storage of previous actions, observations, reasoning traces, and outcomes that may help an agent make better decisions in the future.

The primary research question guiding this project is:

> Can failure-aware structured episodic memory reduce repeated ineffective exploration and improve the performance of autonomous penetration-testing agents?

To study this question, the project will evaluate several memory strategies on multi-step cybersecurity tasks derived from existing benchmarks. Vulnerable environments and Capture-the-Flag (CTF)-style challenges will provide realistic but isolated experimental settings, and agent trajectories, reasoning steps, tool usage, exploit attempts, and outcomes will be logged during execution.

The project will compare three memory settings: (1) No Memory Baseline, in which the agent uses only the current context window without persistent memory; (2) Raw Memory Replay, in which previous actions and observations are appended directly to the context; and (3) Failure-Aware Episodic Memory, in which failed attempts, dead ends, and unsuccessful reasoning traces are summarized and stored in a structured memory format for later retrieval. The evaluation metrics include task success rate, number of repeated failures, token usage, and recovery efficiency.

In the longer term, this work may be extended to study deceptive cybersecurity environments that use honeypots, decoy vulnerabilities, or misleading observations. The failure-aware memory mechanisms may improve agent robustness in these settings and support ongoing work in Dr. Keke Chen's lab on autonomous cybersecurity agents.

Summer Research Plan / Timeline

Period	Planned Work
Weeks 1–2	Literature review on AI agents, episodic memory, and cybersecurity benchmarks
Weeks 2–3	Set up environment, baseline agent, and logging infrastructure
Weeks 3–4	Implement no-memory and raw-memory baselines
Weeks 4–6	Implement failure-aware structured episodic memory mechanisms
Weeks 6–7	Run experiments and collect benchmark data
Weeks 7–8	Analyze results and prepare final report

Anticipated Outcomes

The project will produce (1) an evaluation framework for studying memory mechanisms in cybersecurity agents, (2) experimental datasets containing agent trajectories and failure patterns, and (3) preliminary findings that support future research on deception-aware autonomous cybersecurity agents. All experiments will be conducted in isolated sandbox or Capture-the-Flag environments intended for cybersecurity education and research purposes only.

The project is expected to provide insight into whether structured representations of prior failures can reduce repeated ineffective exploration and improve long-horizon task efficiency in cybersecurity-oriented AI agents. This work may contribute to future research publications or open-source benchmarking tools for evaluating agent memory in cybersecurity applications.

References

[1] A. K. Zhang et al., *Cybench: A Framework for Evaluating Cybersecurity Capabilities and Risks of Language Models*, arXiv:2408.08926, 2024.

[2] X. Shen et al., *PentestAgent: Incorporating LLM Agents for Automated Penetration Testing*, arXiv:2411.05185, 2024.

[3] N. Shinn, F. Cassano, E. Berman, A. Gopinath, K. Narasimhan, and S. Yao, *Reflexion: Language Agents with Verbal Reinforcement Learning*, NeurIPS, 2023.

[4] J. S. Park, J. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, *Generative Agents: Interactive Simulacra of Human Behavior*, UIST, 2023.

[5] K. Nakano et al., *Guided Reasoning in LLM-Driven Penetration Testing Using Structured Attack Trees*, arXiv:2509.07939, 2025.

Shrikant Bhatnagar

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Education

University of Maryland, Baltimore County (UMBC) — Baltimore, MD
B.S. Computer Engineering, Cybersecurity Track — *Expected May 2028*
GPA: 4.0/4.0 *(President's List)*

Skills

Languages: Python, C, C++, Java, React Native, x86 Assembly, Bash, MATLAB
Platforms & Tools: AWS Athena, Arduino IDE, Embedded Linux (Raspberry Pi), React Native, Git, GitHub, Android Studio

Research Experience

Undergraduate Researcher — August 2025 - Present
Future Sensing and Interaction Lab, UMBC — *Baltimore, MD*
Mentor: Dr. Dong Li

- Developed an Android application in React Native to deploy on-device AI models for privacy-preserving physiological signal analysis and mobile health data collection.
- Implemented a real-time, on-device heart-sound acquisition and preprocessing pipeline in React Native, enabling continuous heartbeat detection for machine-learning inference.
- Designed and validated a wired-microphone–based sensing system to capture phonocardiogram (PCG) signals, including peak detection and heart-rate estimation on mobile devices.

National Science Foundation Intern — May 2025 - August 2025
Emerging Software Technologies Lab, UMBC — *Baltimore, MD*
Mentor: Dr. Lei Zhang

- Developed Bash-based cryptographic shell scripts for testing and benchmarking classical and post-quantum cryptographic algorithms aligned with NIST PQC standardization efforts.
- Developed Python-based projects using Qiskit to compare random number generation between classical Python libraries and IBM quantum computer backends, analyzing differences in entropy and quantum randomness.
- Collaborated with a cybersecurity company during the internship to apply post-quantum cryptography concepts and practical security knowledge in real-world environments.

User Interface Developer — June 2024 - August 2024
NanoBios Lab, Indian Institute of Technology Bombay — *Mumbai, India*
Mentor: Dr. Arnab Ghosh

- Designed and developed the user interface for an ambulance tracking application using Flutter.
- Collaborated using GitHub version control, gaining experience in software teamwork and collaborative development workflows.
- Built, tested, and deployed applications for both Android and iOS platforms using Android Studio.

Work Experience

Multidisciplinary Tutor — August 2025 - Present
Academic Success Center, UMBC — *Baltimore, MD*

- Tutored 10+ students weekly through scheduled appointments, along with additional drop-in sessions, in Mathematics, Computer Science, Computer Engineering, and Physics courses.
- Assisted 5+ students weekly with programming projects in Python, C++, and Digital Logic, helping debug code and strengthen foundational problem-solving skills.
- Led collaborative group study sessions for multiple students, providing academic guidance and exam-focused problem-solving strategies to improve quiz and test performance.

Teaching Assistant, Calculus 1 August 2025 - Present
Mathematics Department, UMBC *Baltimore, MD*

- Led 2 discussion sections across 2 semesters for 35+ students in Calculus I, addressing conceptual questions and supporting student learning through guided instruction.
- Held regular office hours and provided academic support to help students strengthen problem-solving skills and improve performance on quizzes and exams.
- Proctored quizzes and exams and assisted in grading 300+ student submissions throughout the semester.

Cybersecurity Intern June 2025 – August 2025
Technuf LLC (NSF Industry Internship) *Rockville, MD*

- Designed and deployed a GenAI-powered log analysis system using Ollama LLaMA-3 LLM with vector search (RAG architecture), processing 1000+ unstructured security logs through a distributed processing pipeline and enabling semantic querying through an AI chatbot interface.
- Built scalable data analysis workflows using Python (NumPy, Pandas), implementing pipelines that made data human-interpretable and supported real-time security monitoring.
- Led knowledge transfer sessions to ensure smooth transition and future maintainance of the project.

Leadership and Projects

Design Build Fly (DBF) Competition Team Member October 2025 – Present
American Institute of Aeronautics and Astronautics (AIAA), UMBC *Baltimore, MD*

- Assisted in MATLAB-based scoring and sensitivity analysis for the aircraft design project, evaluating performance trade-offs and mission optimization strategies.
- Authored and organized a 40+ page technical design report, documenting aircraft anatomy, subsystem design, and engineering justifications for competition submission.
- Participated in the national competition as a test safety pilot, supporting flight testing operations and ensuring safe UAV handling procedures.

hackUMBC Judge & App Developer May 2025 - Present
UMBC Hackathon *Baltimore, MD*

- Developed the official hackUMBC cross-platform mobile application using React Native, supporting 300+ participants for event scheduling and real-time updates with a team of 5 people.
- Evaluated and provided technical feedback on 20 student-built software projects across domains including AI, web, and health systems, assessing architecture, code quality, and innovation.
- Assisted in using AWS Athena to query and analyze participant data stored in S3, supporting real-time response tracking and data-driven decision making for 300+ users.

Autonomous Robotics Development for SUAS Competition October 2024 – Present
American Institute of Aeronautics and Astronautics (AIAA), UMBC *Baltimore, MD*

- Achieved 13th place internationally in the Student Unmanned Aerial Systems (SUAS) Competition as part of the UMBC AIAA autonomous robotics team.
- Engineered computer vision pipelines for an autonomous UAV robotics system, enabling real-time navigation and aerial mapping using Python and OpenCV with GitHub to collaborate between 5 memebers.
- Implemented feature detection and matching algorithms with homography-based alignment to support robust perception and localization from onboard camera data.
- Assisted in developing computer vision–based object detection for precise payload delivery, improving targeting accuracy in autonomous UAV operations.